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# CINEMATCH: AN AI-POWERED MOVIE DISCOVERY & RECOMMENDATION SYSTEM

*A Python Mini-Project Report submitted in partial fulfillment of the requirements for the course of Python Programming (B.Tech AI & DS)*



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## ABSTRACT

With the proliferation of digital streaming content, users are faced with cognitive fatigue when attempting to select relevant entertainment. This project develops **CineMatch**, an intelligent, end-to-end Python web application that solves discoverability challenges using Natural Language Processing (NLP) and mathematical vector similarity modeling. CineMatch unites TF-IDF (Term Frequency-Inverse Document Frequency) text representation with Cosine Similarity metrics and User-Item Collaborative heuristics to synthesize accurate, context-aware movie recommendations.

The application incorporates interactive gamified modules including a Movie Roulette decision engine, a 30-Second Matchmaker Quiz, TMDB CDN poster streaming, and direct YouTube trailer deeplinks. Built with Streamlit and styled with custom responsive CSS, CineMatch is tailored for seamless mobile (iOS/Android) interaction. For global distribution without firewall barriers or subscription fees, the application integrates Cloudflare Quick Tunnels to provide a zero-password HTTPS endpoint alongside a 1-click WhatsApp mobile sharing integration.

## 1. INTRODUCTION & PROBLEM STATEMENT

**1.1 Background:** Traditional categorical navigation fails to account for narrative subtleties, thematic intersections, and fluctuating user moods. Algorithmic recommender systems alleviate choice overload by scoring candidates across multidimensional feature spaces.

**1.2 Project Objectives:**

- Construct an automated NLP feature extraction pipeline using Scikit-Learn.
- Compute pairwise Cosine Similarity across synthetic movie profile documents.
- Engineer a reactive, mobile-first web interface via Python's Streamlit library.
- Implement secure edge distribution using Cloudflare Tunnels and WhatsApp API.

## 2. MATHEMATICAL FOUNDATION & ALGORITHMS

**2.1 TF-IDF Vectorization:** In CineMatch, each movie $d$ in corpus $D$ is transformed into a weighted vector based on Term Frequency (TF) and Inverse Document Frequency (IDF):

$$W(t, d, D) = TF(t, d) \times [\ln((1 + |D|) / (1 + |\{d \in D : t \in d\}|)) + 1]$$

**2.2 Cosine Similarity Metric:** Pairwise similarity between feature vectors $A$ and $B$ evaluates the angular displacement in n-dimensional space, yielding a normalized match score from 0.0 (orthogonal) to 1.0 (identical):

$$\cos(\theta) = (A \cdot B) / (||A|| \times ||B||) = \sum A_i B_i / (\sqrt{\sum A_i^2} \times \sqrt{\sum B_i^2})$$

## 3. SYSTEM ARCHITECTURAL PIPELINE

The system is architected into four decoupled functional tiers:

| Pipeline Stage | Technologies | Description |
|---|---|---|
| **1. Data Ingestion** | Pandas, TMDB CDN | Ingests 30 multi-attribute records; creates synthetic document strings. |
| **2. ML / NLP Engine** | Scikit-Learn (TF-IDF) | Generates sparse vector matrix; computes 30x30 similarity scores. |

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| **3. Reactive UI** | Streamlit, CSS3 | Serves catalog, fuzzy search, Roulette, quiz, and trailer deeplinks. |
| **4. Edge Deployment** | Cloudflare, WhatsApp API | Exposes port 8501 via HTTPS; enables 1-tap mobile WhatsApp sharing. |

## 4. CONCLUSION & REFERENCES

CineMatch demonstrates a robust, production-grade implementation of Python data engineering principles. By bridging mathematical NLP vectorization with high-availability edge tunneling and responsive mobile UX, the project fulfills all academic and functional criteria for modern AI applications.


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